ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 11

ELECTRICAL TECHNOLOGY

MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 11 Electrical Technology practice examination.

Written by Academic Central for practice, following the CAPS curriculum for Grade 11. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Penalise once per question for missing units. Accept other correct answers and small rounding differences.

QUESTION 1

1 1.1 B ✓✓ 1.2 A ✓✓ 1.3 A ✓✓ 1.4 B ✓✓ 1.5 C ✓✓ [10]

QUESTION 2

2.1 An unsafe act is something a person does that can cause an accident ✓, e.g. working on a live circuit or not wearing safety glasses ✓. An unsafe condition is a dangerous state of the workplace or equipment ✓, e.g. a frayed cable, a wet floor or a missing machine guard ✓. (4)

2.2 A live circuit can cause electric shock, burns or death ✓. Any TWO steps: switch off and isolate the circuit at the distribution board ✓; lock the switch and attach a warning tag ✓; test with a tester that the circuit is dead before starting (3)

2.3 Any THREE: muscles contract so the person cannot let go ✓; burns at the entry and exit points ✓; the heart beats irregularly (fibrillation) or stops ✓; breathing stops; falls from a height (3)

[10]

QUESTION 3

3.1 (7)

(a) $V_{max} = 340\ V$ ✓

(b) $\omega = 2\pi f \rightarrow f = 314 \div 2\pi$ ✓ $= 50\ Hz\ (49{,}97\ Hz)$ ✓

(c) $V_{rms} = 0{,}707 \times 340$ ✓ $= 240{,}38\ V$ ✓

(d) $v = 340 \sin(314 \times 0{,}005) = 340 \sin(1{,}57\ rad = 90°)$ ✓ $= 340\ V$ ✓

3.2.1 $X_L = 2\pi fL$ ✓ $= 2\pi \times 50 \times 0{,}2$ ✓ $= 62{,}83\ \Omega$ ✓ (3)

3.2.2 $X_C = 1 \div (2\pi fC)$ ✓ $= 1 \div (2\pi \times 50 \times 100 \times 10^{-6})$ ✓ $= 31{,}83\ \Omega$ ✓ (3)

3.2.3 $Z = \sqrt{(R^2 + (X_L - X_C)^2)}$ ✓ $= \sqrt{(30^2 + 31^2)}$ ✓ $= 43{,}14\ \Omega$ ✓ (3)

3.2.4 $I = V \div Z$ ✓ $= 220 \div 43{,}14$ ✓ $= 5{,}10\ A$ ✓ (3)

3.2.5 $\theta = \cos^{-1}(R \div Z) = \cos^{-1}(30 \div 43{,}14)$ ✓ $= 45{,}94°$ ✓ (3)

The current lags the voltage, because $X_L > X_C$ (the circuit is inductive) ✓

3.2.6 V_L and V_C are 180° out of phase and partly cancel each other ✓. The supply voltage is the phasor sum of V_R, V_L and V_C, so a single component voltage can be larger than the supply ✓. (2)

3.3 $f_r = 1 \div (2\pi\sqrt{(LC)})$ ✓ $= 1 \div (2\pi\sqrt{(0{,}2 \times 100 \times 10^{-6})})$ ✓ $= 35{,}59\ Hz$ ✓ (3)

3.4 The impedance is a minimum and equals R ✓; the current is a maximum ✓; the power factor is 1 (current in phase with voltage) ✓. (3)

[30]

QUESTION 4

4.1 (a) P is connected to positive and N to negative ✓; the depletion layer narrows and current flows once about 0,7 V is exceeded ✓. (4)
(b) P is connected to negative and N to positive ✓; the depletion layer widens and only a tiny leakage current flows (until breakdown) ✓.

4.2 Positive half-cycle: one pair of diagonally opposite diodes (e.g. D1 and D3) conducts ✓ and current flows through the load in one direction ✓. Negative half-cycle: the other pair (D2 and D4) conducts ✓ and current still flows through the load in the same direction ✓, giving full-wave pulsating DC. The capacitor charges at the peaks and discharges into the load between them ✓, which smooths the output (reduces ripple) ✓. (6)

4.3 (8)

(a) $V_R = 15 - 9{,}1 \checkmark = 5{,}9\ V \checkmark$

(b) $I_T = 5{,}9 \div 220 \checkmark = 26{,}82\ mA \checkmark$

(c) $I_L = 9{,}1 \div 1\,000 \checkmark = 9{,}1\ mA \checkmark$

(d) $I_Z = I_T - I_L = 26{,}82 - 9{,}1 \checkmark = 17{,}72\ mA \checkmark$

4.4 $I_C = \beta \times I_B = 100 \times 40\ \mu A \checkmark = 4\ mA \checkmark$ (4)

$I_E = I_C + I_B \checkmark = 4{,}04\ mA \checkmark$

4.5 It limits the current so the LED is not damaged ✓ (3)

$R = (12 - 2) \div 0{,}02 \checkmark = 500\ \Omega \checkmark$

[25]

QUESTION 5

5.1 (5)

$$Y = A(B + \overline{B}) + \overline{A}B \checkmark$$
$$= A \cdot 1 + \overline{A}B \checkmark = A + \overline{A}B \checkmark$$
$$= (A + \overline{A})(A + B) \checkmark = A + B \checkmark$$

5.2 (6)

A	B	C	A + B	$\overline{C}$	Y
0	0	0	0	1	0
0	0	1	0	0	0
0	1	0	1	1	1
0	1	1	1	0	0
1	0	0	1	1	1
1	0	1	1	0	0
1	1	0	1	1	1
1	1	1	1	0	0

All 8 input combinations ✓; A + B column ✓; $\overline{C}$ column ✓; Y column ✓✓✓ (one mark for every two correct rows, rounded down)

5.3 (6)

	$\overline{BC}$	$\overline{B}C$	BC	$B\overline{C}$
$\overline{A}$	0	1	1	0
A	0	1	1	1

Correct map ✓; group of four 1s (m1, m3, m5, m7) gives C ✓✓; pair (m6, m7) gives AB ✓✓

$F = C + AB$ ✓

5.4 The capacitor charges through R_1 and R_2 to ⅔ V_{CC} while the output is high ✓. The comparator then resets the flip-flop, the output goes low and C discharges through R_2 to ⅓ V_{CC} ✓. The cycle repeats on its own, giving a square wave ✓. (5)

$f = 1{,}44 \div ((10\,000 + 2 \times 47\,000) \times 10 \times 10^{-6}) = 1{,}44 \div 1{,}04$ ✓ $= 1{,}38$ Hz ✓

5.5 In a combinational circuit the output depends only on the present inputs ✓. In a sequential circuit the output also depends on the previous state (it has memory) ✓, e.g. a flip-flop, counter or shift register ✓. (3)

[25]

TOTAL: 100

END OF MARKING GUIDELINE